

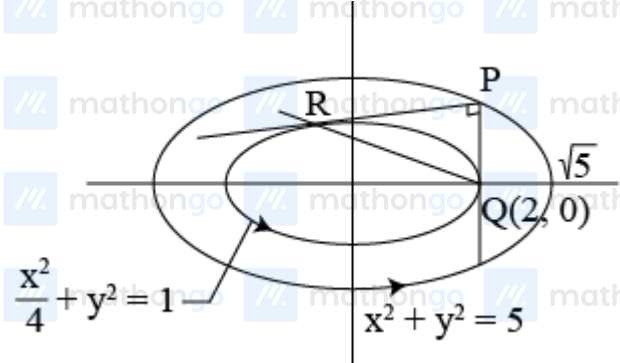
Hints & Solutions

#MathBoleTohMathonGo

Q1 (2) - Single Correct

Here, P lies on $x^2 + y^2 = 5$

Let, $P(\sqrt{5} \cos \theta, \sqrt{5} \sin \theta)$



Comparing chord of contact with middle point

$$\text{and } \frac{xh}{h^2+4k^2} + \frac{yk}{h^2+4k^2} = 1$$

$$\frac{x\sqrt{5} \cos \theta}{4} + \frac{y\sqrt{5} \sin \theta}{1} = 1$$

$$\frac{\frac{\sqrt{5} \cos \theta}{4}}{\frac{h^2+4k^2}{h^2+4k^2}} = \frac{\frac{\sqrt{5} \sin \theta}{1}}{\frac{h^2+4k^2}{h^2+4k^2}} = \frac{1}{1}$$

eliminating θ , we get

$$x^2 + y^2 = \frac{5}{16} (x^2 + 4y^2)^2$$

Q2 (4) - Single Correct

Any normal to the ellipse is

$$\text{i.e. } \frac{\sqrt{5}x}{\cos \theta} - \frac{2y}{\sin \theta} = 1$$

$$y = x \frac{\sqrt{5}}{2} \tan \theta - \frac{\sin \theta}{2}$$

It should be of the form $y = mx + \frac{1}{m}$

$$\Rightarrow m = \frac{\sqrt{5}}{2} \tan \theta, \frac{1}{m} = -\frac{\sin \theta}{2}$$

$$\Rightarrow \frac{\sqrt{5}}{4} \cdot \frac{\sin^2 \theta}{\cos \theta} = -1$$

$$\therefore \cos \theta = -\frac{1}{\sqrt{5}}$$

$$\therefore \theta = \pi - \cos^{-1}\left(\frac{1}{\sqrt{5}}\right) \text{ or } \pi + \cos^{-1}\left(\frac{1}{\sqrt{5}}\right)$$

Q3 (4) - Single Correct

#MathBoleTohMathonGo

Hints & Solutions

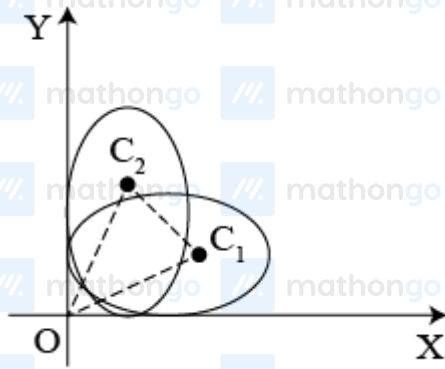
#MathBoleTohMathonGo

Clearly, origin will be a point lying on the director circle of the ellipse.

Thus, equation of the locus of centre of ellipse will be

$$x^2 + y^2 = a^2 + b^2$$

$$\Rightarrow x^2 + y^2 = 100$$



The adjacent figure shows the position of centre of ellipse in two extreme position

$$Q = \angle C_1OC_2$$

$$= \frac{\pi}{2} - 2 \tan^{-1} \left(\frac{5}{5\sqrt{3}} \right)$$

$$= \frac{\pi}{6}$$

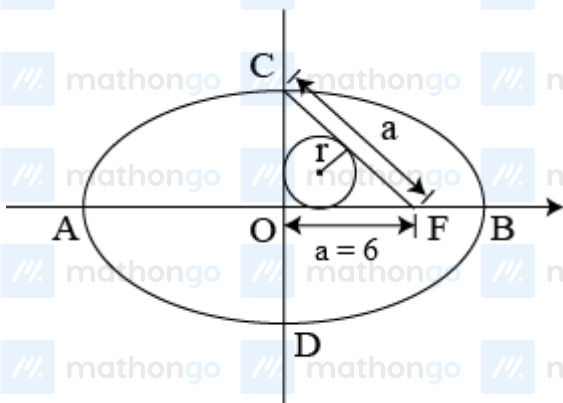
$$\text{Thus, arc length } C_1C_2 = 10 \cdot \frac{\pi}{6} = \frac{5\pi}{3}$$

Q4 (1) - Single Correct

$$a^2e^2 = 36 \Rightarrow a^2 - b^2 = 36 \dots(i)$$

$$\text{using } r = (s - a) \tan \frac{A}{2} \text{ in } \triangle OCF$$

$$\Rightarrow 1 = (s - a) \tan 45^\circ, \text{ when } a = CF$$



#MathBoleTohMathonGo

Hints & Solutions

#MathBoleTohMathonGo

$$\Rightarrow 2 = 2(s - a)$$

$$\therefore 2 = 2s - 2a$$

$$2 = 2s - AB = (OF + OC + CF) - AB$$

$$\Rightarrow 2 = 6 + \frac{CD}{2} + \frac{AB}{2} - AB$$

$$\Rightarrow \frac{AB - CD}{2} = 4$$

$$\Rightarrow 2(a - b) = 8$$

$$\Rightarrow a - b = 4 \quad \dots (ii)$$

$$\text{from Eqs. (i) and (ii), } a + b = 9 \dots (iii)$$

$$\therefore 2a = 13, 2b = 5$$

$$\Rightarrow (AB)(CD) = 65$$

Q5 (3) - Single Correct

If PSQ is focal chord then $\frac{1}{SP} + \frac{1}{SQ} = \frac{2}{l}$, l = semilatus rectum.

The product of perpendiculars from foci to any tangent is b^2 .

The points A, B lie on auxiliary circle,

$$\therefore \frac{SP+SQ}{SP \cdot SQ} = \frac{2a}{b^2} = \frac{8}{9}$$

$$\text{and } SA \cdot S'B = 9, OA \cdot OB = a^2 = 16$$

$$\Rightarrow \frac{SP+SQ}{SP \cdot SQ} + \frac{OA \cdot OB}{SA \cdot S'B} = \frac{8}{9} + \frac{16}{9} = \frac{24}{9}$$

Q6 (2) - Single Correct

$$\frac{ax}{\cos \theta} - \frac{by}{\sin \theta} = (a^2 - b^2)$$

$$\frac{ah}{\cos \theta} = a^2 - b^2 \quad [\text{as normals pass through } (h, 0)]$$

$$\Rightarrow \cos \theta = \frac{ah}{a^2 - b^2}$$

$$\Rightarrow \left| \frac{ah}{a^2 - b^2} \right| \leq 1 \text{ or } |h| \leq he^2$$

Q7 (2) - Single Correct

Circle C is $x^2 + y^2 - 16 = 0$, it intersects the circle $x^2 + y^2 - 16x - 10y + 9 = 0$, orthogonally

$$\text{Then, } \lambda = \frac{7}{5}$$

$$\text{Now, } |r_1 - r_2| < C_1 C_2 < r_1 + r_2$$

$\therefore$ Number of common tangents is 2.

#MathBoleTohMathonGo

Hints & Solutions

#MathBoleTohMathonGo

Q8 (3) - Single Correct

Using property $S_1P_i \times S_2Q_i = b^2$

$$\Rightarrow 20b^2 = 5120$$

$$\Rightarrow b = 16 \text{ and } a = 20$$

$$\therefore e = \frac{3}{5}$$

Q9 (3) - Single Correct

Let standard equation of ellipse be $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ and hyperbola be $\frac{x^2}{a_1^2} - \frac{y^2}{b_1^2} = 1$

We have, major axis = transverse axis

$$2a = 2a_1$$

$$\Rightarrow a = a_1$$

and radius of director circle of ellipse

$$a^2 + b^2 = 4R^2$$

and radius of director circle of hyperbola

$$a_1^2 - b_1^2 = R^2$$

On dividing Eqs. (ii) by (iii), we get

$$\frac{a^2 + b^2}{a_1^2 - b_1^2} = 4$$

$$\Rightarrow a^2 + b^2 = 4a_1^2 - 4b_1^2$$

$$\Rightarrow b^2 = 3a^2 - 4b_1^2 \quad [\text{from Eq.(i)}]$$

$$\text{Or } b_1^2 = \frac{1}{4}(3a^2 - b^2)$$

$$\text{We have, } 1 - e_1^2 = \frac{b^2}{a^2}$$

Now, on dividing Eqs. (v) by (vi), we get

$$\frac{1 - e_1^2}{e_2^2 - 1} = \frac{b^2}{b_1^2}$$

$$\Rightarrow \frac{1 - e_1^2}{e_2^2 - 1} = \frac{4b^2}{3a^2 - b^2}$$

$$\Rightarrow \frac{4}{3\frac{a^2}{b^2} - 1} = \frac{3}{1 - e_1^2} - 1$$

$$\Rightarrow \frac{1 - e_1^2}{e_2^2 - 1} = \frac{4(1 - e_1^2)}{3 - 1 + e_1^2}$$

$$\Rightarrow e_1^2 + 2 = 4(e_2^2 - 1)$$

$$\therefore 4e_2^2 - e_1^2 = 6$$

#MathBoleTohMathonGo

Hints & Solutions

#MathBoleTohMathonGo

Q10 (1) - Single Correct

We have, the curve $y^2 - x^2 = 1$ Its abscissa = n and ordinate = $\sqrt{1+n^2}$

$\therefore P$ is $(n, \sqrt{1+n^2})$

d_n = Shortest distance = Perpendicular distance of the line

$$y = x$$

$$\Rightarrow dx = \frac{\sqrt{1+n^2}-n}{\sqrt{2}} = \lim_{n \rightarrow \infty} \frac{n(\sqrt{1+n^2}-n)}{\sqrt{2}} \times \frac{\sqrt{1+n^2}+n}{\sqrt{1+n^2}+n}$$

$$\therefore \lim_{n \rightarrow \infty} \frac{n}{\sqrt{2}[\sqrt{1+n^2}+n]} = \frac{1}{2\sqrt{2}}$$

Q11 (4) - Single Correct

We have, the hyperbola

$$\frac{x^2}{16} - \frac{y^2}{9} = 1$$

and eccentricity $e = \frac{5}{4}$

$\therefore$ Foci are (5,0) and (-5,0)

$\therefore$ Foci are (5,0) and (-5,0)

Clearly, the given line $3x + (5 - 4\sqrt{2})y = 15$ passes through the one focus (5,0)

Hence, the reflected ray must be passes through other focus (-5,0)

and clearly only option (d) is passes through (-5,0).

$\therefore$ Reflected ray is $3x - y(4\sqrt{2} + 5) + 15 = 0$

Q12 (3) - Single Correct

Let $y = f(x)$ be a conic and (x_1, y_1) be a point on the conic such that the segment of the tangent between the coordinate axes is bisected by the point.

$\therefore$ Equation of tangent at (x_1, y_1) $y - y_1 = p(x - x_1)$,

where $p = \left(\frac{dy}{dx}\right)_{(x_1, y_1)}$

$A = X$ -intercept = $\left(x_1 - \frac{y_1}{p}, 0\right)$

$B = Y$ -intercept = $(0, y_1 - px_1)$

Mid-point of $AB = \left(\frac{x_1 - \frac{y_1}{p}}{2}, \frac{y_1 - px_1}{2}\right) = (x_1, y_1)$

#MathBoleTohMathonGo

Hints & Solutions

#MathBoleTohMathonGo

$$\Rightarrow x_1 - \frac{y_1}{p} = 2x_1 \text{ and } \frac{y_1 - px_1}{2} = y_1$$

$$\Rightarrow \frac{y_1}{p} = -x_1 \text{ and } -px_1 = y_1$$

$$\therefore p = -\frac{y_1}{x_1} \text{ and } p = -\frac{y_1}{x_1}$$

$$\therefore \left(\frac{dy}{dx}\right)_{(x_1, y_1)} = -\frac{y_1}{x_1}$$

$$\Rightarrow \frac{dy}{dx} = -\frac{y}{x}$$

$$\Rightarrow \int \frac{1}{y} dy = -\int \frac{1}{x} dx + C$$

$$\Rightarrow \ln y = -\ln x + \ln C$$

$$\Rightarrow y = \frac{C}{x}$$

$$\therefore xy = C$$

Since, this curve passes through (2,4) $xy = 8$ is equation of conic. $xy = (2\sqrt{2})^2$ is rectangular hyperbola having eccentricity $= \sqrt{2}$

We know that, foci of the rectangular hyperbola is $(\pm C\sqrt{2} \pm C\sqrt{2})$

$\therefore (4, 4)$ and $(-4, -4)$ are two foci of the conic.

Q13 (2) - Single Correct

Let the hyperbola is $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

$$\therefore e = \sqrt{1 + \frac{b^2}{a^2}} = \frac{\sqrt{a^2 + b^2}}{a}$$

$$\text{and } e' = \frac{\sqrt{a^2 + b^2}}{b}$$

Equation of line is $\frac{x}{e} + \frac{y}{e'} = 1$

$$\text{i.e. } \frac{ax}{\sqrt{a^2 + b^2}} + \frac{by}{\sqrt{a^2 + b^2}} = 1$$

$$\text{If } \frac{a}{\sqrt{a^2 + b^2}} = \cos \theta, \frac{b}{\sqrt{a^2 + b^2}} = \sin \theta$$

$$x \cos \theta + y \sin \theta = 1$$

Since, this line touching fixed circle.

$$\text{Let fixed circle be } x^2 + y^2 = r^2$$

$$\therefore r = 1$$

$$\therefore \text{Area of circle} = \pi r^2 = \pi$$

Q14 (1) - Single Correct

#MathBoleTohMathonGo

Hints & Solutions

#MathBoleTohMathonGo

Given, $P(x, y)$ lies on $x^2 + y^2 = 1$

Let $P = (\cos \theta, \sin \theta)$ and $(x + y)^2 \leq \lambda$

$$\Rightarrow x^2 + y^2 + 2xy < \lambda$$

$$\Rightarrow 2xy \leq \lambda - 1$$

$$\Rightarrow 2 \sin \theta \cos \theta \leq \lambda - 1$$

$$\Rightarrow \sin 2\theta \leq \lambda - 1 \quad [\because |\sin \theta| \leq 1]$$

$$\therefore -1 \leq \lambda - 1 \leq 1$$

$$\Rightarrow 0 \leq \lambda \leq 2$$

Now, check the position of $(\lambda, 0)$ w.r.t. the hyperbola,

$$(x - 2)^2 - y^2 = 1$$

$$(\lambda - 2)^2 - 1 = \text{negative}$$

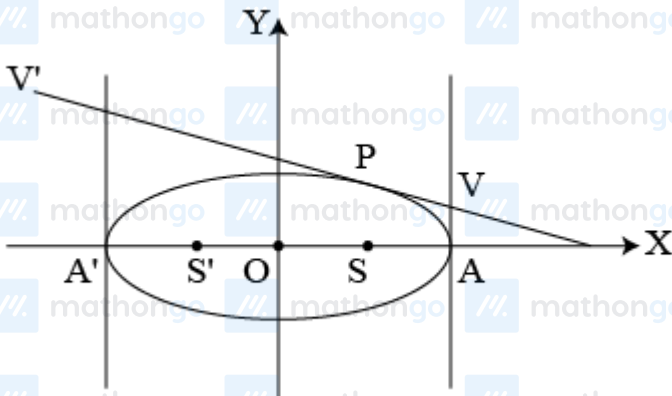
Since, $(\lambda, 0)$ lies inside the hyperbola

Hence, no tangent can be drawn.

Q15 (2, 3, 4) - Multiple Correct

Any tangent at $P(a \cos \theta, b \sin \theta)$ is

$$\frac{x}{a} \cos \theta + \frac{y}{b} \sin \theta = 1 \dots (i)$$



line AV is $x = a$ and $A'V'$ is $x = -a$

$$\text{So points } V = \left(a, \frac{b(1 - \cos \theta)}{\sin \theta} \right)$$

$$\text{and } V' = \left(-a, \frac{b(1 + \cos \theta)}{\sin \theta} \right)$$

$$\text{So, } I(AV) \cdot I(A'V') = \frac{b(1 - \cos \theta)}{\sin \theta} \cdot \frac{b(1 + \cos \theta)}{\sin \theta} = b^2$$

$$\text{Slope of line } V'S \text{ is } = \frac{b(1 + \cos \theta)}{\sin \theta(-a - ae)} = m_1$$

#MathBoleTohMathonGo

Hints & Solutions

#MathBoleTohMathonGo

also, slope of VS is $= \frac{b(1-\cos\theta)}{\sin\theta(a-ae)} = m_2$

$$\text{Now, } m_1 m_2 = \frac{b^2(1-\cos^2\theta)}{\sin^2\theta(-b^2)} = -1$$

$\Rightarrow \angle V'SV = 90^\circ$ and $V'S'VS$ is a cyclic quadrilateral.

Q16 (1, 2) - Multiple Correct

$$h = \pm ae, k = \pm \frac{b^2}{a}$$

$$\therefore k = \pm a(1 - e^2) = \pm a\left(1 - \frac{h^2}{a^2}\right) = \pm\left(a - \frac{h^2}{a}\right)$$

On taking + ve sign, we get

$$k = a - \frac{h^2}{a} \Rightarrow h^2 = a(a - k) \dots(i)$$

$\therefore$ locus of (h, k) is $x^2 = a(a - y)$

On taking - ve sign, we get

$$k = -a + \frac{h^2}{a}$$

$$\Rightarrow h^2 = a(a + k) \dots(ii)$$

$\therefore$ locus of (h, k) , is $x^2 = a(a + y)$

Q17 (1, 2, 3) - Multiple Correct

If PSS' form a Δ , then ellipse

$$PS + PS' > SS'$$

$$\Rightarrow K \in (-2, 2)$$

If PSS' are collinear the pair of coincident lines $\Rightarrow K = \pm 2$ If PSS' neither form triangle nor collinear

$$\text{Then } PS + PS' < SS'$$

$$\Rightarrow k \in (-\infty, -2) \cup (2, \infty)$$

Q18 (1, 2, 3) - Multiple Correct

Focal property of ellipse

$$PS + PS' = 2a, a > b$$

$$PS + PS' = 2b, b > a$$

$$PS \cos \alpha + PS' \cos \beta = 2ae \dots(i)$$

#MathBoleTohMathonGo

Hints & Solutions

#MathBoleTohMathonGo

$$PS \sin \alpha + PS' \sin \beta = 0 \dots(ii)$$

$$PS + PS' = 2a \dots(iii)$$

from Eqs. (i) and (ii), we get

$$PS = \frac{2ae \cdot \sin \beta}{\sin(\alpha + \beta)}, PS' = \frac{2ae \cdot \sin \alpha}{\sin(\alpha + \beta)} \dots(iv)$$

from Eqs. (iii) and (iv)

$$e(\sin \alpha + \sin \beta) = \sin(\alpha + \beta)$$

$$\therefore e \left[2 \sin \left(\frac{\alpha + \beta}{2} \right) \cdot \cos \left(\frac{\alpha - \beta}{2} \right) \right] = 2 \sin \left(\frac{\alpha + \beta}{2} \right) \cdot \cos \left(\frac{\alpha + \beta}{2} \right)$$

$$\therefore e \left(\cos \frac{\alpha}{2} \cdot \cos \frac{\beta}{2} + \sin \frac{\alpha}{2} \cdot \sin \frac{\beta}{2} \right) = \cos \frac{\alpha}{2} \cdot \cos \frac{\beta}{2} - \sin \frac{\alpha}{2} \cdot \sin \frac{\beta}{2}$$

$$\Rightarrow \tan \frac{\alpha}{2} \cdot \tan \frac{\beta}{2} = \frac{1-e}{1+e}$$

Q19 (2, 4) - Multiple Correct

Clearly (α, β) on curve will be of the form $\left(\frac{2m^2-1}{2m^2+1}, \frac{-2m}{2m^2+1} \right)$

Q20 (1, 3, 4) - Multiple Correct

Verify each option in $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ or $\frac{x^2}{a^2} - \frac{y^2}{b^2} = -1$ and $xy = c^2$

$$\frac{2x}{a} = t + \frac{1}{t} \text{ and } \frac{2y}{b} = t - \frac{1}{t}$$

$$\frac{x}{a} + \frac{y}{b} = t \dots(i)$$

$$\text{and } \frac{x}{a} - \frac{y}{b} = \frac{1}{t} \dots(ii)$$

On multiplying Eqs. (i) and (ii), we get

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

Which is a hyperbola.

$$(2) \frac{t}{a} - \frac{y}{b} = 1 \dots(i)$$

$$\text{and } \frac{x}{a} + \frac{yt}{b} - 1 = 0 \dots(ii)$$

On solving Eqs. (i) and (ii), we get

$$\frac{x}{a} = \frac{1-t^2}{1+t^2} \text{ and } \frac{y}{b} = \frac{2t}{1+t^2}$$

which is not a hyperbola.

$$(3) x = e^t + e^{-t} \dots(i)$$

$$\text{and } y = e^t - e^{-t} \dots(ii)$$

#MathBoleTohMathonGo

Hints & Solutions

#MathBoleTohMathonGo

$$\Rightarrow x + y = 2e^t \text{ and } x - y = 2e^{-t}$$

$$\therefore (x + y)(x - y) = 4$$

$$\Rightarrow x^2 - y^2 = 4$$

which is a hyperbola.

$$(4) x^2 - 6 = 2 \cos t \dots(i)$$

$$\Rightarrow y^2 + 2 = 4 \cos^2 \frac{t}{2} = 2(1 + \cos t)$$

$$\therefore y^2 = 2 \cos t \dots(ii)$$

$$\Rightarrow x^2 - y^2 - 6 = 0$$

$$\Rightarrow x^2 - y^2 = 6$$

Which is a hyperbola.

Q21 (1, 2, 4) - Multiple Correct

We have,

$$(px^2 + qy^2 + r)(4x^2 + 4y^2 - 8x - 4) = 0$$

$$S_1 \equiv px^2 + qy^2 + r = 0$$

$$\text{and } S_2 \equiv 4x^2 + 4y^2 - 8x - 4 = 0$$

(1) S_2 represents a circle and if $r = 0$ and p and q are of the opposite signs, then S_1 represents a pair of straight lines.

(2) Clearly, if $p = q$ and r is negative, then S_1 represents a circle. S_2 is a circle.

(3) When p and q are of opposite sign $r \neq 0$, then S_1 represents a hyperbola not an ellipse.

(4) When p and q are of same sign but unequal and r is of sign opposite to that of p . S_1 represents an ellipse. S_2 represents a circle.

Q22 (1, 2, 4) - Multiple Correct

$$\text{We have, } (1 - x^2) \frac{dy}{dx} + xy = ax$$

$$\frac{dy}{dx} + \frac{x}{1-x^2} \cdot y = \frac{ax}{1-x^2}$$

$$\text{Now, } IF = e^{\int \frac{x}{1-x^2} dx} = e^{-\frac{1}{2} \log(1-x^2)} = (1-x^2)^{-\frac{1}{2}} = \frac{1}{\sqrt{1-x^2}}$$

$$\Rightarrow y \cdot \frac{1}{\sqrt{1-x^2}} = \int \frac{ax}{(1-x^2)^{3/2}} dx$$

$$\text{Let } 1 - x^2 = t^2, \text{ then } -x dx = t dt$$

Hints & Solutions

#MathBoleTohMathonGo

$$\Rightarrow \frac{y}{\sqrt{1-x^2}} = -a \int \frac{t}{t^3} dt = -a \int \frac{1}{t^2} dt = -\frac{at^{-1}}{-1} = \frac{a}{1}$$

$$\therefore \frac{y}{\sqrt{1-x^2}} = \frac{a}{\sqrt{1-x^2}} + C$$

$$\therefore y = a + c\sqrt{1-x^2}$$

$$\Rightarrow (y-a)^2 = c^2(1-x^2)$$

$$\Rightarrow c^2x^2 + (y-a)^2 = c^2$$

$$\Rightarrow x^2 + \frac{(y-a)^2}{c^2} = 1$$

(1) Eq. (i) represents a conic which is either an ellipse or a hyperbola. Principle axes parallel to coordinate axes.

(2) Centre = $(0, a)$ and length of the one principle axis is 2.

Q23 (3, 4) - Multiple Correct

We have, $a \sec \theta - x \tan \theta - y = 0 \dots(i)$

and $b \sec \theta + y \tan \theta - x = 0 \dots(ii)$

By cross-multiplication, we get

$$\frac{\sec \theta}{x^2+y^2} = \frac{\tan \theta}{-ax+by} = \frac{1}{ay+bx}$$

$$\Rightarrow \sec \theta = \frac{x^2+y^2}{ay+bx} \text{ and } \tan \theta = \frac{by-ax}{ay+bx}$$

$$\therefore \sec^2 \theta - \tan^2 \theta = 1$$

$$\Rightarrow \frac{(x^2+y^2)^2}{(ay+bx)^2} - \frac{(by-ax)^2}{(ay+bx)^2} = 1$$

$$\Rightarrow (x^2+y^2)^2 - (by-ax)^2 = (ay+bx)^2$$

$$\Rightarrow x^4 + y^4 + 2x^2y^2 - b^2y^2 - a^2x^2 + 2abxy = a^2y^2 + b^2x^2 + 2abxy$$

$$\Rightarrow (x^2+y^2)^2 - b^2(x^2+y^2) - a^2(x^2+y^2) = 0$$

$$\Rightarrow x^2 + y^2 - a^2 - b^2 = 0$$

$$\Rightarrow x^2 + y^2 = a^2 + b^2$$

It is the director circle of $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

and director circle of circle $x^2 + y^2 = \frac{a^2+b^2}{2}$.

Q24 (2, 3) - Multiple Correct

We have, the hyperbola $x^2 - 3y^2 = 3$

$$\Rightarrow \frac{x^2}{3} - y^2 = 1$$

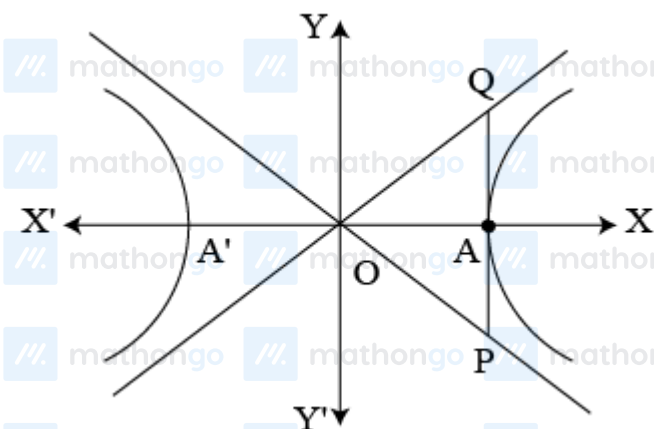
#MathBoleTohMathonGo

Hints & Solutions

#MathBoleTohMathonGo

Equation of tangent at $(\sqrt{3}, 0)$ to the given hyperbola is

$$x = \sqrt{3}$$



and asymptotes of the hyperbola

$$\Rightarrow \frac{x}{\sqrt{3}} \pm y = 0$$

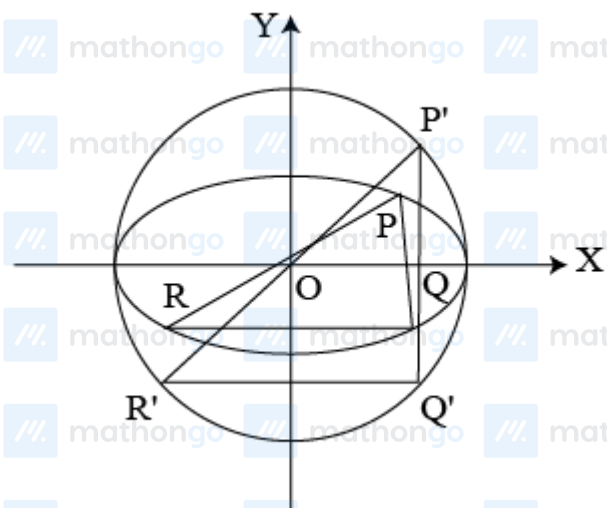
$$\therefore y = \pm \frac{x}{\sqrt{3}}$$

Clearly, $\triangle OPQ$ forms an equilateral triangle, then

$$\text{its area} = \frac{\sqrt{3}}{4} \times (2)^2 = \frac{\sqrt{3} \times 4}{4} = \sqrt{3} \text{ sq units}$$

Q25 (1) - Multiple Correct

$$\frac{\text{ar} \cdot \triangle PQR}{\text{ar} \cdot \triangle P'Q'R'} = \frac{\sqrt{5}}{3}$$



$$\Rightarrow \frac{b}{a} = \frac{\sqrt{5}}{3}$$

$$e^2 = 1 - \frac{b^2}{a^2}$$

#MathBoleTohMathonGo

Hints & Solutions

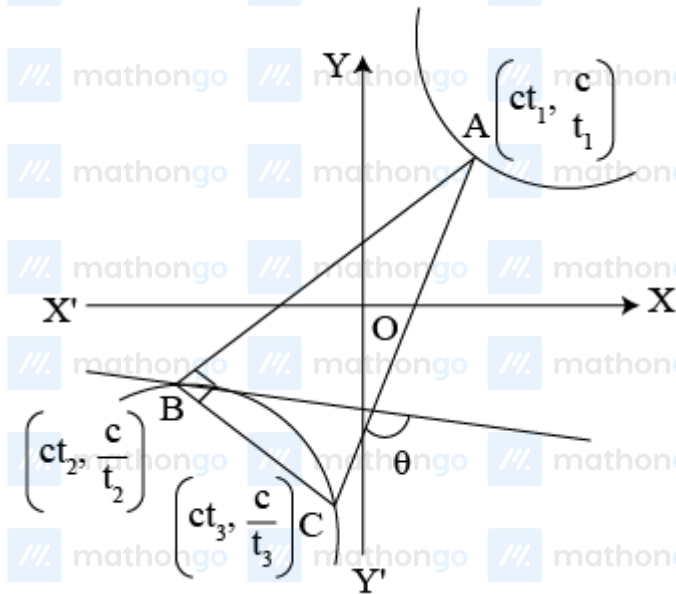
#MathBoleTohMathonGo

$$\Rightarrow e^2 = 1 - \frac{5}{9}$$

$$\Rightarrow e = \frac{2}{3}$$

Q26 (4) - Multiple Correct

Now, the slope of tangent at $B \left(ct_2, \frac{c}{t_2} \right)$ is $-\frac{1}{t_2^2}$, and the slope of chord joining A and C is $-\frac{1}{t_1 t_3}$.



We have, $AB \perp BC$

$$\Rightarrow -\frac{1}{t_1 t_2} \times -\frac{1}{t_2 t_3} = -1$$

$$\Rightarrow t_1 t_2^2 t_3 = -1 \dots (i)$$

Let θ be the angle between the tangent at B and AC

$$\text{i.e. } \tan \theta = \frac{-\frac{1}{t_1 t_3} + \frac{1}{t_2^2}}{1 + \frac{1}{t_2^2 t_1 t_3}}$$

From Eq. (i), we get $\tan \theta = \infty$

$$\therefore \theta = 90^\circ$$

Q27 (3) - Multiple Correct

Equation of normal at $\left(ct, \frac{c}{t} \right)$ to hyperbola $xy = c^2$

$$xt^3 - yt - ct^4 + c = 0 \dots (i)$$

It touches its conjugate.

$$\text{Equation of tangent, } x + yt^2 - 2ct = 0 \dots (ii)$$

#MathBoleTohMathonGo

Hints & Solutions

#MathBoleTohMathonGo

Since, Eqs. (i) and (ii) are identical, then

$$\frac{t_3}{1} = \frac{-t}{t^2} = \frac{c-ct^4}{2ct} = \frac{1-t^4}{2t}$$

$$\Rightarrow 2t^4 = 1 - t^4$$

$$\Rightarrow 3t^4 = 1 \text{ [it is 4 degree]}$$

So, there are four normals.

Q28 (3) - Paragraph 1

Explanation (For Question 29, 30)

$$2x + 3y + 9 = 0$$

$$P(-1, 2) \text{ and } Q(-2, 3)$$

$$\text{Hence, image of point } P \text{ form line } 2x + 3y + 9 = 0 \Rightarrow \frac{x+1}{2} = \frac{y-2}{3} = \frac{-2(-2+6+9)}{4+9}$$

$$\Rightarrow x = -5, y = -4$$

$$\Rightarrow P'(-5, -4)$$

$$\text{Now length of } P'Q = \sqrt{9 + 49} = \sqrt{58} = 2a$$

$$\Rightarrow a = \frac{\sqrt{58}}{2}$$

$$\text{We know that, } 2ae = PQ \Rightarrow ae = \frac{1}{\sqrt{2}}$$

$$\therefore a^2 e^2 = \frac{1}{2}$$

$$b^2 = a^2 - a^2 e^2 = \frac{58}{4} - \frac{1}{2} = \frac{56}{4}$$

$$b^2 = 14$$

$$\Rightarrow \text{length of minor axis} = 2\sqrt{14}$$

$$\text{Area of ellipse} = \pi ab = \pi \frac{\sqrt{58}}{2} \sqrt{14} = \pi \sqrt{203}$$

Equation of line $P'Q$ is

$$\Rightarrow (y - 3) = \frac{4-3}{-3}(x + 2)$$

$$\Rightarrow 3y - 9 = 7x + 14$$

$$\text{or } 7x - 3y + 23 = 0$$

On solving $7x - 3y + 23 = 0$ and $2x + 3y + 9 = 0$, we get

$$A\left(-\frac{32}{9}, -\frac{17}{27}\right)$$

Q29 (4) - Paragraph 1

#MathBoleTohMathonGo

Q30 (2) - Paragraph 2

Explanation (For Question 31, 32)

$$\text{Here, } b^2 = \left| \frac{2+1-9}{\sqrt{2}} \right| \left| \frac{4+1-9}{\sqrt{2}} \right| = 12$$

$$2ae = 2 \Rightarrow ae = 1$$

$$b^2 = a^2 - a^2 e^2 \Rightarrow a^2 = 13$$

$$\Rightarrow e = \frac{1}{\sqrt{13}}$$

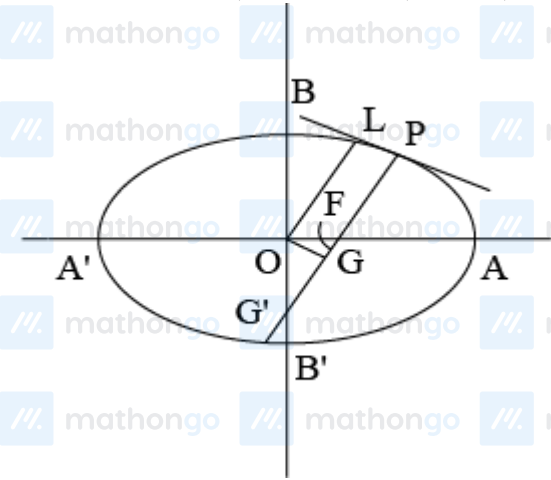
$$\text{length of major axis} = 2\sqrt{13} = \sqrt{52}$$

$$\text{length of latus-rectum} = \frac{2b^2}{a} = \frac{2 \times 12}{\sqrt{13}} = \frac{24}{\sqrt{13}}$$

Q31 (3) - Paragraph 2

Q32 (1) - Paragraph 3

Let P be $(a \cos \theta, b \sin \theta)$. Normal at P intersect the major and minor axes at G and G' . Then, the coordinates of G and G' are $\left(\frac{a^2-b^2}{a} \cos \theta, 0\right)$ and $\left(0, -\frac{a^2-b^2}{a} \sin \theta\right)$ respectively.



$$\begin{aligned} \therefore PG^2 &= \left(a \cos \theta - \frac{a^2 - b^2}{a} \cos \theta \right)^2 + b^2 \sin^2 \theta \\ &= \frac{b^4 \cos^2 \theta}{a^2} + b^2 \sin^2 \theta \end{aligned}$$

$$= \frac{b^2}{a^2} (b^2 \cos^2 \theta + a^2 \sin^2 \theta)$$

$$\text{Similarly, } PG'^2 = \frac{a^2}{b^2} (b^2 \cos^2 \theta + a^2 \sin^2 \theta)$$

Again, OF is perpendicular from O on the normal and OL is perpendicular from C on the tangent,

$$\frac{x}{a} \cos \theta + \frac{y}{b} \sin \theta = 1$$

Hints & Solutions

#MathBoleTohMathonGo

Clearly,

$$PF = OL = \frac{1}{\sqrt{\frac{\cos^2 \theta}{a^2} + \frac{\sin^2 \theta}{b^2}}}$$

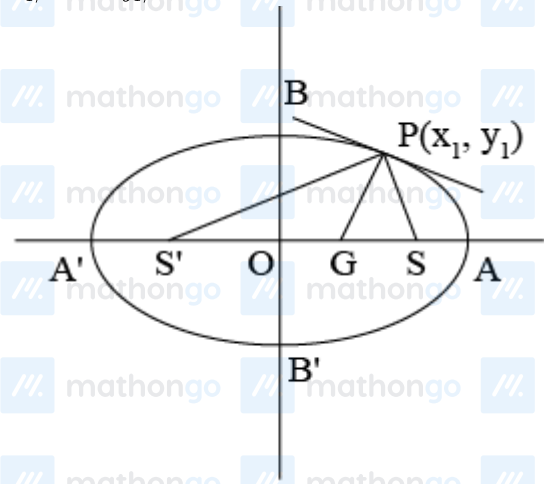
$$= \frac{ab}{\sqrt{b^2 \cos^2 \theta + a^2 \sin^2 \theta}}$$

$$\text{Then, } PF \cdot PG = \frac{ab}{\sqrt{b^2 \cos^2 \theta + a^2 \sin^2 \theta}} \cdot \frac{b}{a} \sqrt{b^2 \cos^2 \theta + a^2 \sin^2 \theta} = b^2$$

Q33 (3) - Paragraph 3

Let P be the point (x_1, y_1) the normal at which is

$$\frac{x-x_1}{x_1/a^2} = \frac{y-y_1}{y_1/b^2}$$



It meets the major axis $Y = 0$ at point

$$G \left[x_1 \left(1 - \frac{b^2}{a^2} \right), 0 \right] \text{ or } (e^2 x_1, 0) \quad \left[\because e^2 = 1 - \frac{b^2}{a^2} \right]$$

$$\therefore OG = e^2 x_1$$

$$\therefore S'G = S'O + OG = ae + e^2 x_1 = e(a + ex_1) = e \cdot SP$$

$$\text{and } SG = OS - OG = ae - e^2 x_1 = e(a - ex_1) = e \cdot SP$$

$$\therefore \frac{S'G}{SG} = \frac{e \cdot SP}{e \cdot SP} = \frac{S'P}{SP}$$

Q34 (4) - Paragraph 4

We have, rectangular hyperbola

$$xy - y - 2x - 2 = 0 \dots(i)$$

On differentiating w.r.t. x , we get

Hints & Solutions

#MathBoleTohMathonGo

$$xy' + y' - y' - 2 = 0$$

$$\Rightarrow y' = \frac{2-y}{x-1}$$

Slope of tangent at (5,3) is $-\frac{1}{4}$

Slope of normal at (5,3) is 4

$\therefore$ Equation of normal is $y - 3 = 4(x - 5)$

$$y = 4x - 17 \dots (ii)$$

Since, this normal intersect again at hyperbola

$$x(4x - 17) - 4x + 17 - 2x - 2 = 0$$

$$\Rightarrow 4x^2 - 23x + 15 = 0$$

$$\Rightarrow (x - 5)(4x - 3) = 0$$

$$\Rightarrow x = \frac{3}{4} \text{ and } x = 5$$

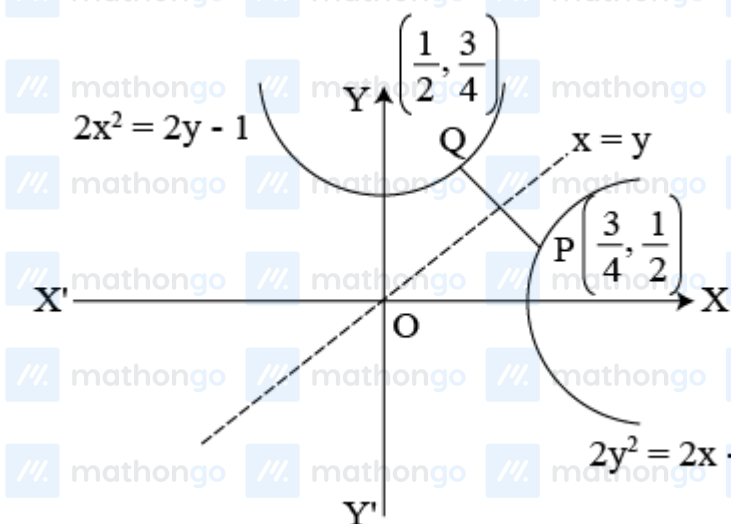
$\therefore$ Point is $\left(\frac{3}{4}, -14\right)$.

Q35 (2) - Paragraph 4

Clearly, given parabolas $2y^2 = 2x - 1$ and $2x^2 = 2y - 1$ are

symmetrical about the line $y = x$.

$\therefore$ Shortest distance occurs along the common normal which is perpendicular to the line $y = x$



On differentiating $2y^2 = 2x - 1$ w.r.t. x , we get

Hints & Solutions

#MathBoleTohMathonGo

$$2 \cdot 2y \frac{dy}{dx} = 2$$

$$\therefore \frac{dy}{dx} = \frac{1}{2y} = 1$$

$$\Rightarrow y = \frac{1}{2}$$

Now, the points are

$$P\left(\frac{3}{4}, \frac{1}{2}\right) \text{ and } Q\left(\frac{1}{2}, \frac{3}{4}\right)$$

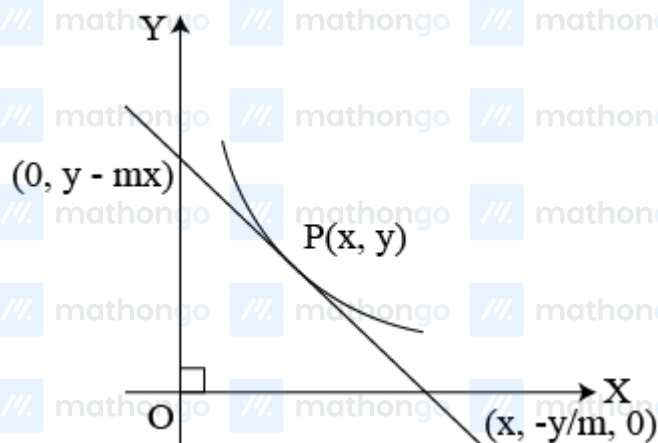
$$\text{Hence, the shortest distance} = \sqrt{\frac{1}{16} + \frac{1}{16}} = \frac{1}{2\sqrt{2}}$$

Q36 (2) - Paragraph 5

Explanation (For Question 37, 38)

$Y - y = m(X - x)$, if $Y = 0$. Then, $X = x - \frac{y}{m}$ and if $X = 0$, then $Y = y - mx$

$$\text{Hence, } x - \frac{y}{m} = 2x$$



$$\Rightarrow \frac{dy}{dx} = -\frac{y}{x}$$

$$\int \frac{dy}{y} + \int \frac{dx}{x} = c$$

$$\Rightarrow xy = c$$

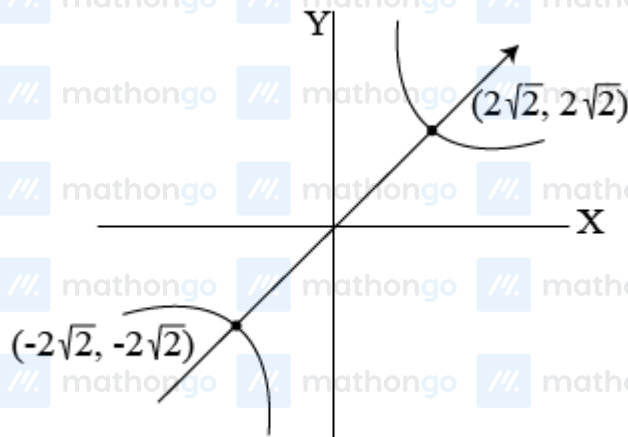
Passes through (2,4)

$\Rightarrow$ Equation of conic is $xy = 8$ which is a rectangular hyperbola with $a = \sqrt{2}$. Hence, the two vertices are

$(2\sqrt{2}, 2\sqrt{2})$ and $(-2\sqrt{2}, -2\sqrt{2})$. Foci are (4,4) and (-4,-4)

Hints & Solutions

#MathBoleTohMathonGo



$\therefore$ Equation of S is $x^2 + y^2 = 32$

Q37 (4) - Paragraph 5

Q38 (2) - Integer Type

Here, $(2\alpha, \alpha)$ has between $x^2 + y^2 = 9 + 3$ and $\frac{x^2}{9} + \frac{y^2}{3} = 1$

$$\Rightarrow 5\alpha^2 - 12 < 0 \text{ and } \frac{4\alpha^2}{9} + \frac{\alpha^2}{3} - 1 > 0$$

$$\Rightarrow \frac{9}{7} < \alpha^2 < \frac{12}{5}$$

$\therefore$ possible integral value of $\alpha^2 = 2$

Q39 (4) - Integer Type

The director circle of $\frac{x^2}{2} + \frac{y^2}{1} = 1$, is $x^2 + y^2 = 3$

Let, $P(\sqrt{3} \cos \theta, \sqrt{3} \sin \theta)$ chord of contact is

$$x(\sqrt{3} \cos \theta) + y(2\sqrt{3} \sin \theta) = 2$$

It touches $x^2 + y^2 = r^2$

$$\Rightarrow r = \frac{2}{\sqrt{3 \cos^2 \theta + 12 \sin^2 \theta}} = \frac{2}{\sqrt{3 + 9 \sin^2 \theta}}$$

$$r_{\max} = \frac{2}{\sqrt{3}}, r_{\min} = \frac{2}{\sqrt{12}}$$

$$\text{So, } \frac{A_{\max}}{A_{\min}} = 4$$

Q40 (0) - Integer Type

#MathBoleTohMathonGo

Hints & Solutions

#MathBoleTohMathonGo

We have,

$$\frac{x^2}{14} + \frac{y^2}{5} = 1 \dots (i)$$

The equation of normal to this ellipse at $P(\sqrt{14} \cos \theta, \sqrt{5} \sin \theta)$ is $\sqrt{14} \sec \theta x - \sqrt{5} \operatorname{cosec} \theta y = 9$

This meets the ellipse (i) at $Q(\sqrt{14} \cos 2\theta, \sqrt{5} \sin 2\theta)$

$$\therefore 14 \sec \theta \cos 2\theta - 5 \operatorname{cosec} \theta \sin 2\theta = 9$$

$$\Rightarrow 18 \cos^2 \theta - 9 \cos \theta - 14 = 0$$

$$\Rightarrow (6 \cos \theta - 7)(3 \cos \theta + 2) = 0$$

$$\Rightarrow 3 \cos \theta + 2 = 0$$

Q41 (4) - Integer Type

For the ellipse

$$\frac{x^2}{16} + \frac{y^2}{9} = 1$$

the equation of director circle is $x^2 + y^2 = 25$.

The director circle will cut the ellipse $\frac{x^2}{50} + \frac{y^2}{20} = 1$ at four points.

Hence, the number of points is 4.

Q42 (3) - Integer Type

Here, $y = \frac{x}{2} + 2$ is tangent on the ellipse

$$\text{Then, } 4 = 4 \left(\frac{1}{4} \right) + b^2$$

$$\Rightarrow b^2 = 3$$

Q43 (2) - Integer Type

We have $\frac{(x-4)^2}{16} + \frac{(y-3)^2}{9} = 1$ any point P on the curve is $P[4(1 + \cos \theta), 3(1 + \sin \theta)]$

The reflection of the point P about the line $x - y - 2 = 0$ can be found using

$$\begin{aligned} \frac{h - 4(1 + \cos \theta)}{1} &= \frac{k - 3(1 + \sin \theta)}{-1} \\ &= \frac{-2[4(1 + \cos \theta) - 3(1 + \sin \theta) - 2]}{2} \end{aligned}$$

$$\therefore h = 4(1 + \cos \theta) + 4 \cos \theta - 3 \sin \theta - 1$$

$$\Rightarrow h = 2 + 4 \cos \theta$$

#MathBoleTohMathonGo

Hints & Solutions

#MathBoleTohMathonGo

$$\text{and } k = 3(1 + \sin \theta) - 4 \cos \theta + 3 \sin \theta + 1$$

$$\Rightarrow k = 5 + 4 \sin \theta$$

$$\therefore \left(\frac{h-5}{3}\right)^2 + \left(\frac{k-2}{4}\right)^2 = 1$$

$$\therefore \text{Equation of image is } 16(x-5)^2 + 9(y-2)^2 = 144$$

$$\Rightarrow 16x^2 + 9y^2 - 160x - 36y + 292 = 0$$

On comparing with given equation, we get

$$k_1 = -160, k_2 = 292$$

$$\therefore k_1 + k_2 = 132 \Rightarrow \text{units digit} = 2.$$

Q44 (8) - Integer Type

We have, the equation of asymptotes is

$$xy + 2x + 4y + c = 0$$

On comparing it with

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + k = 0$$

$$a = 0, h = \frac{1}{2}, b = 0, g = 1, f = 2 \text{ and } k = c$$

It represents a pair of straight lines, if

$$abk - 2fgh - af^2 - bg^2 - ch^2 = 0$$

$$\Rightarrow 0 - 2(2)(1)\left(\frac{1}{2}\right) - 0 - 0 - c\frac{1}{4} = 0$$

$$\Rightarrow -2 - \frac{c}{4} = 0 \Rightarrow \frac{c}{4} = -2$$

$$\therefore c = 8$$

Q45 (2) - Integer Type

We have, $\frac{e}{2}$ and $\frac{e'}{2}$ are eccentricities of a hyperbola and its conjugate hyperbola.

$$\therefore \frac{4}{e^2} + \frac{4}{e'^2} = 1$$

$$\Rightarrow 4 = \frac{e^2 e'^2}{e^2 + e'^2}$$

The line passes through the point $(e, 0)$ and $(0, e')$.

$$\frac{x}{e} + \frac{y}{e'} = 1$$

$$\Rightarrow e'x + ey - ee' = 0$$

It is tangent to the circle

Hints & Solutions

#MathBoleTohMathonGo

$$x^2 + y^2 = r^2$$

$$\therefore r = \frac{ee'}{\sqrt{e^2 + e'^2}}$$

$$\Rightarrow r^2 = \frac{e^2 e'^2}{e^2 + e'^2} = 4$$

$$\therefore r = 2$$

Q46 (4) - Integer Type

Clearly, $C(0, 0)$ let the equation of circle be

$$x^2 + y^2 - ax - by = 0$$

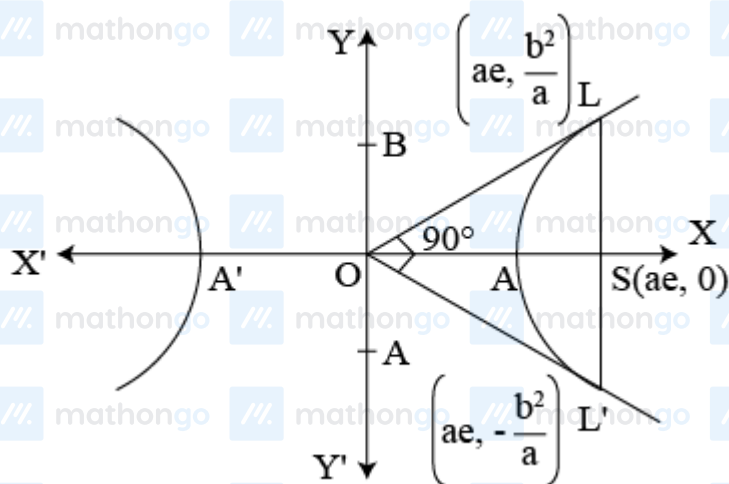
On homogenising with chord $11x + 3y - 14 = 0$ and comparing with pair of asymptotes, we get equation of

$$\text{circle as } x^2 + y^2 - \frac{112}{23}x - \frac{64}{23}y = 0$$

$$\therefore \frac{23(a - b + c)}{12} = \frac{23\left(+\frac{112}{23} - \frac{64}{23}\right)}{12} = 4$$

Q47 (1) - Integer Type

Clearly, $\angle SOL = 45^\circ$



$$\text{In } \triangle OSL, \tan 45^\circ = \frac{\frac{b^2}{a}}{ae} = 1$$

$$\therefore \frac{b^2}{a^2 e} = 1 \Rightarrow e = \frac{b^2}{a^2} = e^2 - 1$$

$$\Rightarrow e^2 - e - 1 = 0 \Rightarrow e = \frac{1 + \sqrt{1+4}}{2}$$

$$\therefore e = \frac{\sqrt{5}+1}{2} \text{ i.e. } [e] = 1$$

Hints & Solutions

#MathBoleTohMathonGo

Q48 (4) - Integer Type

$$x = \frac{55-y}{2y-5}, y = \frac{55+5x}{2x+1}$$

Satisfied by 16 pairs of (x, y)

$$\therefore \lambda = 16$$

$$[\sqrt{\lambda}] = 4$$